Spencer Willett

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Experience

- **Freelance Software Engineer** — San Diego, CA
  Software Engineer — *Sept 2024 - present*
  - **NVIDIA Jetson Dev Kit**: Implemented secure boot for AI startup
  - **Wifi and BLE**: Created and tested wifi and BLE connection methods for the NVIDIA Jetson Dev Kit
  - **AI Webform Automator**: Created a chrome extension for a client to upload a PDF file and automatically fill in a webform
- **Qualcomm** — San Diego, CA
  Software Engineer - Secure Systems Group — *May 2022 - Dec 2023*
  - **C Library**: Built a library in C for use on Embedded Linux and Android. Involved concepts used for IPC.
  - **Trusted Virtual Machine Development**: Integrated the C Library I built and updated the unit testing suite used to test the Trusted Virtual Machine
  - **Test Driven Development**: Designed C++ code with a Test Driven Development approach using the Google Test framework
  - **Static Analysis**: Fixed code security vulnerabilities for multiple repositories using an internal static analysis tool to audit code
  - **Open Source**: Contributed actively to both open-source and proprietary code bases
  - **Gerrit/GitHub**: Used both Gerrit and GitHub for software collaboration and code reviews
- **Intel** — Chandler, AZ
  Software Engineer - Graduate Student Intern — *Nov 2019 - Dec 2021*
  - **12x Optimization**: Designed and implemented a 12x data processing optimization (using multithreaded code) to reduce the run time from 1 hour to under 5 minutes. (C#)
  - **Group Presentation**: Gave a presentation to a group of software engineers on development tools and potential uses for the group: Docker, GitLab CI/CD
  - **Docker**: Designed and implemented a replica database using Docker to use for development (previously the group has used the production database during testing phases)
- **Arizona State University** — Tempe, AZ
  Graduate Student Assistant - CSE 539 Applied Cryptography — *Aug 2021 - Dec 2021*
  - **Course Design**: Proposed, designed, and implemented extra credit projects for the students: Time-Based One-Time Passwords (TOTP) in Python to match the Google Authenticator app on their phones.
  - **Projects**: Developed correct solutions for the five major projects in the course: Steganography, cryptanalysis, MD5 hash collisions, Diffie-Hellman key exchange, and RSA key exchange (C#)

Education

- **Arizona State University** — Tempe, AZ
  Master of Science in Computer Science — *Aug. 2019 – Dec. 2021*
- **Brigham Young University** — Provo, UT
  Bachelor of Science in Physics — *2017*

Projects

- **Virtual Computer**: I built a virtual computer following the Nand2Tetris course offered by the Hebrew University of Jerusalem

Honors/Awards

- **Hackathon 2nd Place Team**: I was the lead software engineer on my team. I built a basic LeNet convolutional neural network using Keras in Python. Hosted by FOX at ASU.

Programming Skills

- **Languages**: C++, C, Python, Rust — **Technologies**: Linux, NVIDIA Jetson Dev Kit, GDB